

Riemann Hypothesis via Self-Dual Trace Confinement: A Conditional Closure

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Abstract

For any positive integer weights m_ρ , the Riemann hypothesis is equivalent to the vanishing of $A_Z(\zeta) = \sum_\rho m_\rho (\operatorname{Re} \rho - \frac{1}{2})^2$ over the nontrivial zeros. A companion paper proves a duality-confinement theorem: for a strongly measurable observable with values in a real Hilbert space that intertwines an involution with a linear isometric involution and has square-integrable odd part, the Gram operator of the odd part vanishes whenever it is dominated by positive trace-class operators of vanishing trace. This paper applies the theorem to the reflection $s \mapsto 1 - \bar{s}$ in the critical line and measures what the application is worth. A named hypothesis $\Gamma_{\text{SDTC-Selberg}}$ on a shell of represented zeros, supplying domination records and a bridge to the zeros' coordinates, implies that every represented zero has coordinate $\frac{1}{2}$; together with a map from coordinates to the reals that sends $\frac{1}{2}$ to $\frac{1}{2}$ and carries the represented zeros onto the real parts of all nontrivial zeros, it implies the Riemann hypothesis. Conversely, the hypothesis holds exactly when every represented zero has coordinate $\frac{1}{2}$, so on the bare shell of all nontrivial zeros with their real parts it is equivalent to the Riemann hypothesis; so is the concrete version on finite windows of actual zeros that are closed under $s \mapsto 1 - \bar{s}$, carry unit weights and eventually contain every zero. The conditional theorem is therefore a reformulation, not evidence. By elementary linear algebra we also show that, for windows closed under $s \mapsto 1 - s$ with symmetric positive weights that eventually contain every zero, several natural bounds for the window sums are again equivalent to the Riemann hypothesis, and that on a nonempty window operator-level transport bounds for the identity target cannot tend to zero. No arithmetic estimate is supplied, and no unconditional result is claimed. The mathematical statements are formalized in Lean 4 with Mathlib, using its statement of the Riemann hypothesis; Appendix A gives the scope.

Keywords. Riemann hypothesis; zeros of the zeta function; conditional theorem; positive operators; self-dual trace confinement; Six Birds Theory; Lean formalization.

1 Introduction

The Riemann hypothesis [19] asserts that every nontrivial zero ρ of the Riemann zeta function satisfies $\operatorname{Re} \rho = \frac{1}{2}$. Because the nontrivial zeros are the zeros of the completed zeta function $\Lambda_\zeta(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ in the strip $0 < \operatorname{Re} s < 1$, the hypothesis can be restated as the vanishing of a single nonnegative quantity,

$$A_Z(\zeta) = \sum_\rho m_\rho \left(\operatorname{Re} \rho - \frac{1}{2} \right)^2 \in [0, \infty],$$

the sum running over the nontrivial zeros ρ with positive weights m_ρ (for instance their multiplicities). This reformulation is elementary (Theorem 5.1); its interest here is that it places the Riemann

hypothesis inside the setting of a companion paper [29]. There J is an involution of a measure space X , ψ is a strongly measurable observable with values in a real Hilbert space that intertwines J with a linear isometric involution, and its odd part ψ_- is square-integrable and vanishes exactly on the fixed locus of J . The Gram operator $A_X = \int_X \psi_- \psi_-^*$ is then shown to vanish—and the mass to lie on the fixed locus—as soon as it is dominated by positive trace-class operators of vanishing trace; the proof is a one-line trace squeeze. The global square-integrability hypothesis would require $A_Z(\zeta) < \infty$, which is not known to hold unconditionally; the applications below use either an abstract form of the theorem or finite windows of zeros, where no such issue arises. For the zeros of ζ the relevant involution is

$$J_L(s) = 1 - \bar{s},$$

reflection in the critical line, which maps zeros to zeros by the functional equation and the reflection principle; its fixed locus is the critical line, and the odd readout is $\psi_-(s) = \operatorname{Re} s - \frac{1}{2}$.

This paper examines that route carefully. Its conclusions are as follows.

1. *A conditional theorem.* A named hypothesis $\Gamma_{\text{SDTC-Selberg}}$ (Hypothesis 6.1) supplies positive domination records of vanishing trace for an auxiliary duality-confinement ledger, together with a bridge encoding, at each represented zero ρ of a shell (Section 4), the displacement coordinate $c(\rho) - \frac{1}{2}$ in that ledger’s odd readout. Under this hypothesis every represented zero has coordinate $c(\rho) = \frac{1}{2}$ (Theorem 7.2). If, in addition, a map from coordinates to real numbers, sending $\frac{1}{2}$ to $\frac{1}{2}$, carries the represented zeros onto the real parts of all nontrivial zeros, then the Riemann hypothesis holds (Theorem 7.3).
2. *A calibration of the hypothesis.* For every shell, the hypothesis holds exactly when every represented zero has coordinate $\frac{1}{2}$; a one-point ledger suffices (Proposition 6.2). On the bare shell, whose represented zeros are the nontrivial zeros with their real parts as coordinates, the hypothesis is therefore equivalent to the Riemann hypothesis. The same is true of the operator form, with the actual zeros as objects, in which the duality-confinement theorem is applied to a sequence of finite windows of actual zeros, each closed under J_L , that eventually contains every zero (Theorem 7.4). The conditional theorem is therefore a reformulation of the Riemann hypothesis in the language of domination records, not evidence for it.
3. *Finite windows.* On a finite window W of zeros closed under the functional equation, with weights invariant under it, the window sum $\sum_{\rho \in W} m_\rho (\operatorname{Re} \rho - \frac{1}{2})^2$ is the squared norm of a vector that is invisible to the natural invariant probe (Section 9). This identifies several “payments”—upper bounds for the window sums, or for operators dominating them, that one might hope to establish independently. These facts are elementary. Along windows that eventually contain every zero, each of several natural payment conditions is again equivalent to the Riemann hypothesis; and on a nonempty window, operator-level transport bounds for the identity target, of the form made precise in Proposition 9.2, cannot tend to zero.

No unconditional statement about the zeros of ζ is claimed. Two features should be said at the outset. First, no quantitative or prime-side input is used or supplied beyond the classical zero-free half-plane. The arguments use only analytic facts about ζ —the functional equation and conjugation symmetry, countability of the zeros, the poles and non-vanishing of the Gamma factor, and the absence of zeros with $\operatorname{Re} s \geq 1$ —and for any countable set of points in the strip with positive weights and the same symmetries they prove the corresponding statements, with that set’s own critical-line property in place of the Riemann hypothesis. Second, the functional-analytic content of the companion theorem is not exercised: the conditional theorem uses an abstract trace squeeze, and the concrete form uses operators on $\mathbb{R}$, that is, real numbers. What the paper contributes is a

precise account of what a proof along these lines would have to supply, and a demonstration that the obvious candidates are no easier than the Riemann hypothesis itself. The classical programs aimed at the Riemann hypothesis—spectral interpretations [2], trace formulas in noncommutative geometry [5], Hilbert spaces of entire functions [7, 8], and Weil’s positivity criterion [31, 4] and Li’s criterion [14]—are not used, and nothing here offers an alternative to them.

The mathematical statements of Sections 3 to 7 and 9 are formalized in Lean 4 with the Mathlib library, and the Riemann hypothesis in the conclusions is Mathlib’s statement of it; the records of Section 8 are formalized as lists of labels, and the remarks are not formalized (Appendix A). The conditional theorem is stated for a *shell*, a record whose fields name the data an intended trace-formula realization would provide (Section 4); beyond the represented zeros and their coordinates these data carry no analytic content.

Organization. Section 2 explains the framework vocabulary. Section 3 defines the involution, the readout and the zero ledgers; Section 4 describes the shell; Section 5 proves that the ledger vanishes exactly when the zeros lie on the critical line. Sections 6 and 7 state the hypothesis $\Gamma_{\text{SDTC-Selberg}}$, prove the conditional theorems and their calibration. Section 8 records a classification of the argument’s hypotheses in the audit vocabulary of a further companion paper. Section 9 treats finite windows. Sections 10 to 12 discuss scope and context.

2 Framework

This section explains the framework terms used in the paper. No proof depends on them, and a reader interested only in the mathematics may skip to Section 3.

The author’s Six Birds framework [24] describes a mathematical theory as the collection of fixed points of an idempotent completion rule; later papers add a discipline for which candidate objects are admissible [25] and a finite vocabulary for what an observer can see and certify [26]. Within it, a *structural law* is a named hypothesis asserting that a specified class of structures carries certain data, and the data it asserts are called its *recognition source*. The point of the naming is bookkeeping: every result that uses the hypothesis can be seen to use it.

The structural law relevant here is Self-Dual Trace Confinement (SDTC) of the companion paper [29], in the setting recalled in Section 1 (a strongly measurable observable in a real Hilbert space intertwining an involution with a linear isometric involution): in a specified class of closures with a genuine involutive self-duality, observed through a separating odd readout, the structure supplies positive trace-class operators B_n with $A_X \preceq B_n$ and $\text{tr } B_n \rightarrow 0$, where A_X is the Gram operator of the square-integrable odd readout. That paper proves the corresponding confinement theorem and its converse, and shows that for a single ledger with a separating readout the hypothesis is equivalent to confinement. The present paper specializes the hypothesis to the zeros of the zeta function, where it is called $\Gamma_{\text{SDTC-Selberg}}$: the subscript records that its intended source is a trace formula of Selberg type, which is not constructed here. Consistently with the companion paper, $\Gamma_{\text{SDTC-Selberg}}$ turns out to be equivalent to the Riemann hypothesis on the bare shell of all nontrivial zeros with their real parts (Sections 6 and 7).

Two further companion papers supply vocabulary used in Sections 8 and 9: an audit calculus for classifying the hypotheses of an argument [23], and a residual operator measuring what a probe fails to see [30]. Only elementary special cases of the latter are used, and they are defined where needed.

3 Involution and ledger

This section introduces the functional-equation involution, its odd readout, and the two zero ledgers. Everything here is elementary; the point is to place the critical line in the pattern of an involution, its fixed locus, and a readout that changes sign.

Definition 3.1 (Functional-equation involution J_L). The functional-equation involution is the map $J_L : \mathbb{C} \rightarrow \mathbb{C}$,

$$J_L(s) = 1 - \bar{s}.$$

If $s = x + iy$, then $J_L(s) = (1 - x) + iy$: the map is reflection in the vertical line $\operatorname{Re} s = \frac{1}{2}$.

Directly from the formula, $J_L(J_L(s)) = 1 - \overline{1 - \bar{s}} = s$, and $J_L(s) = s$ if and only if $\operatorname{Re} s = \frac{1}{2}$. Thus J_L is an involution whose fixed locus is the critical line. The completed zeta function satisfies $\Lambda_\zeta(1 - s) = \Lambda_\zeta(s)$ and $\Lambda_\zeta(\bar{s}) = \overline{\Lambda_\zeta(s)}$ [22, 9], so J_L maps zeros of Λ_ζ to zeros of Λ_ζ , preserving the strip $0 < \operatorname{Re} s < 1$.

Definition 3.2 (Anti-invariant readout ψ_-). The anti-invariant readout of J_L is the signed horizontal displacement from the critical line,

$$\psi_-(s) = \operatorname{Re} s - \frac{1}{2}.$$

Since $\operatorname{Re} J_L(s) = 1 - \operatorname{Re} s$, the readout is odd: $\psi_-(J_L(s)) = -\psi_-(s)$. It vanishes exactly on the fixed locus of J_L , so it is separating in the sense of the companion paper.

Definition 3.3 (Nontrivial-zero ledger). The *nontrivial zeros* are the zeros ρ of Λ_ζ with $0 < \operatorname{Re} \rho < 1$; they are exactly the zeros of ζ in that strip. A *nontrivial-zero ledger* is the set Z_ζ^{nt} of nontrivial zeros together with positive integer weights m_ρ ($\rho \in Z_\zeta^{\text{nt}}$). The natural choice of m_ρ is the multiplicity of ρ as a zero of Λ_ζ ; every result below holds for arbitrary positive integer weights, and the formal development uses unit weights where a choice is needed.

Definition 3.4 (Anti-invariant zero ledger). The *anti-invariant zero ledger* is the extended nonnegative sum

$$A_Z(\zeta) = \sum_{\rho \in Z_\zeta^{\text{nt}}} m_\rho |\psi_-(\rho)|^2 = \sum_{\rho \in Z_\zeta^{\text{nt}}} m_\rho \left(\operatorname{Re} \rho - \frac{1}{2} \right)^2 \in [0, \infty],$$

defined as the supremum of its finite partial sums.

No convergence assumption is needed: the value $+\infty$ is allowed, and it is only the vanishing of $A_Z(\zeta)$ that matters. Because every summand is nonnegative and every weight is positive, $A_Z(\zeta) = 0$ exactly when every nontrivial zero lies on the critical line; Section 5 states this as Theorem T. For a finite set $W \subseteq Z_\zeta^{\text{nt}}$, the *window sum* $A_W = \sum_{\rho \in W} m_\rho (\operatorname{Re} \rho - \frac{1}{2})^2$ is a real number bounded by $A_Z(\zeta)$; windows are the subject of Section 9.

4 The saturated Selberg trace shell

The conditional theorem of Section 7 is stated for a record that collects, under one name, the data an intended trace-formula realization of the zeros would provide. We call it the saturated completed Selberg trace shell and write $\operatorname{Sel}_{\zeta, \text{tr}}^!$. The name records the intended realization, through a trace formula of Selberg type [20, 10, 12]. Spectral realizations of the zeros of ζ exist—Connes’ trace formula [5] and Meyer’s spectral interpretation [15], which realizes Weil’s explicit formula [31] as a trace formula—but none is known to come with a self-adjoint operator H such that every nontrivial

zero is $\frac{1}{2} + i\lambda$ with λ in the spectrum of H , which would prove the Riemann hypothesis. (For the Selberg zeta function of a compact hyperbolic surface, self-adjointness of the Laplacian places the zeros coming from eigenvalues $\lambda \geq \frac{1}{4}$ on the critical line, while eigenvalues in $(0, \frac{1}{4})$ give finitely many real exceptional zeros.) Nothing of this kind is constructed or used here.

Definition 4.1 (Saturated completed Selberg trace shell). A *saturated completed Selberg trace shell* $\text{Sel}_{\zeta, \text{tr}}^!$ consists of the following data:

- (1) sets H_L , I_{tr} , EQ_{tr} and EM_{zero} , naming a history carrier, a trace instrument, trace observables and verifier events;
- (2) sets Q_L and M_L with a map $\pi_L : M_L \rightarrow Q_L$, naming a current and a predictive quotient and their comparison;
- (3) a *coordinate system*: a set R with elements 0 and $\frac{1}{2}$ and operations $x \mapsto -x$, $(x, y) \mapsto x - y$ and $x \mapsto 1 - x$, such that $x \mapsto 1 - x$ is an involution whose only fixed point is $\frac{1}{2}$, $x - \frac{1}{2} = 0$ if and only if $x = \frac{1}{2}$, and $(1 - x) - \frac{1}{2} = -(x - \frac{1}{2})$; the *points* of the shell are pairs $s = (x(s), y(s)) \in R^2$, and the *coordinate* of a point s is $x(s)$;
- (4) the functional-equation involution $J_L(s) = (1 - x(s), y(s))$ on points, whose fixed points are those with $x(s) = \frac{1}{2}$, a set V with a subset of *zero values*, and a function Λ_L from points to V (the intended completed zeta function);
- (5) a zero ledger: an element $1 \in R$, a relation $<$ on R (no laws are imposed on it), a set of *represented zeros* with an injective map $\rho \mapsto \text{point}$, and positive integer weights m_ρ ; we write $c(\rho)$ for the coordinate of the point of ρ . It is required that every represented zero satisfies $0 < c(\rho) < 1$ and $\Lambda_L(\rho)$ is a zero value, and every point s with $0 < x(s) < 1$ and $\Lambda_L(s)$ a zero value is represented;
- (6) a function $q : R \rightarrow [0, \infty]$ vanishing exactly at 0 (the intended square), and the anti-invariant ledger $A_Z(\zeta) = \sum_\rho m_\rho q(c(\rho) - \frac{1}{2})$ of the represented zeros, an extended nonnegative sum;
- (7) a visibility relation Vis_L between histories and instruments, and an audit marker Audit_L .

Of these data, the proofs in this paper use only the represented zeros, their weights and coordinates, the law “ $x - \frac{1}{2} = 0$ if and only if $x = \frac{1}{2}$ ”, and q . The remaining laws and data describe the intended realization and serve to specify the bare shell below; the operation $x \mapsto 1 - x$ is a formal symbol subject only to the laws listed. Items (1), (2) and (7) are names: the shell imposes no laws on them, and in particular it does not assert that the trace instrument satisfies a trace formula, that the quotients have any universal property, or that the shell is admissible in the sense of the framework’s formation rules [25]; the audit marker records a status, not a proof of admissibility. A shell is useful precisely because it separates the zeros it represents from the actual zero set: the conditional theorem speaks about represented zeros and their coordinates, and Theorem 7.3 adds the hypothesis that a map from coordinates to the reals, sending $\frac{1}{2}$ to $\frac{1}{2}$, carries them onto the real parts of all nontrivial zeros. One shell is always available: the *bare shell*, with $R = \mathbb{R}$ and its usual operations, element 1 and order, points $\mathbb{C}$, $V = \mathbb{C}$ with zero value 0, $\Lambda_L = \Lambda_\zeta$, represented zeros exactly the nontrivial zeros of Definition 3.3 with unit weights, $q(x) = x^2$, and trivial documentary fields (1), (2) and (7). For it $c(\rho) = \text{Re } \rho$ and $A_Z(\zeta)$ is the ledger of Definition 3.4.

5 Translation theorem T

The next theorem translates the vanishing of the anti-invariant ledger into the critical-line statement. It is a positivity argument. We state it for the nontrivial-zero ledger of Definition 3.3; the identical argument applies to the ledger of represented zeros of any shell, which is how it is used in Section 7.

Theorem 5.1 (Translation theorem T). *For any positive weights m_ρ ,*

$$A_Z(\zeta) = 0 \iff \operatorname{Re} \rho = \frac{1}{2} \text{ for every } \rho \in Z_\zeta^{\text{nt}}.$$

Consequently $A_Z(\zeta) = 0$ if and only if the Riemann hypothesis holds.

Proof. The two directions are Theorems 5.2 and 5.3. The nontrivial zeros are exactly the zeros of ζ in the strip $0 < \operatorname{Re} s < 1$, because $\pi^{-s/2}\Gamma(s/2)$ has neither zeros nor poles there; and every zero of ζ other than the trivial zeros $-2, -4, \dots$ lies in that strip. For the latter, the classical facts used are [22]: ζ has no zeros with $\operatorname{Re} s \geq 1$; $\zeta(0) = -\frac{1}{2}$; $\Lambda_\zeta(s) = \Lambda_\zeta(1-s)$; and $\pi^{-s/2}\Gamma(s/2)$ never vanishes and is finite except at $s = 0, -2, -4, \dots$. If $\zeta(s) = 0$ with $\operatorname{Re} s \leq 0$, then $s \neq 0$, and if s is not a trivial zero the factor at s is finite and nonzero, so $\Lambda_\zeta(s) = 0$, hence $\Lambda_\zeta(1-s) = 0$; since $\operatorname{Re}(1-s) \geq 1$ and $1-s \neq 1$, the factor at $1-s$ is finite and nonzero, so $\zeta(1-s) = 0$, a contradiction. Hence the critical-line statement on Z_ζ^{nt} is the Riemann hypothesis. $\square$

Theorem 5.2 (Translation theorem T, forward direction). *If $A_Z(\zeta) = 0$, then $\operatorname{Re} \rho = \frac{1}{2}$ for every $\rho \in Z_\zeta^{\text{nt}}$.*

Proof. Each summand $m_\rho(\operatorname{Re} \rho - \frac{1}{2})^2$ is nonnegative and is at most the extended sum $A_Z(\zeta)$. If $A_Z(\zeta) = 0$, each summand is therefore zero, and since $m_\rho > 0$ we get $(\operatorname{Re} \rho - \frac{1}{2})^2 = 0$. $\square$

Theorem 5.3 (Translation theorem T, reverse direction). *If $\operatorname{Re} \rho = \frac{1}{2}$ for every $\rho \in Z_\zeta^{\text{nt}}$, then $A_Z(\zeta) = 0$.*

Proof. Every summand vanishes, so every partial sum vanishes, and so does their supremum. $\square$

The equivalence between $A_Z(\zeta) = 0$ and the critical-line statement uses nothing about the zeros beyond positivity of the weights; only its identification with the Riemann hypothesis uses the analytic facts recalled in the proof. Its role is to let the rest of the paper work with the single quantity $A_Z(\zeta)$, or with its finite window sums, instead of with the zero set.

6 The recognition source

By Theorem T, the Riemann hypothesis is the statement $A_Z(\zeta) = 0$. The duality-confinement route asks for this vanishing to come from domination records of vanishing trace. The following hypothesis states what the conditional theorem of Section 7 consumes; the formal record carries in addition some structure that no proof uses (weights, a response involution, a quantitative separation modulus), listed in Appendix A. It is phrased with an auxiliary ledger in the abstract form of the duality-confinement theorem [29, Remark 7.2], because the formal statement allows the domination records to live in any ordered set with a trace, not only among operators.

Hypothesis 6.1 (The recognition source $\Gamma_{\text{SDTC-Selberg}}$). Let $\text{Sel}_{\zeta, \text{tr}}^!$ be a shell. We say that the recognition source $\Gamma_{\text{SDTC-Selberg}}$ *holds for* $\text{Sel}_{\zeta, \text{tr}}^!$ if there exist the following data.

- (i) *An auxiliary ledger.* A finite set X with an involution J (the map $\rho \mapsto x_\rho$ of item (iv) need not be injective), a set of *visible* objects, a response set Y with a zero element, and a readout $\psi_- : X \rightarrow Y$ that is separating on visible objects: for visible x , $\psi_-(x) = 0$ if and only if $Jx = x$.
- (ii) *An ordered set with a trace.* A set C with a relation $\preceq$, a subset of positive elements, a zero element, a distinguished positive element A_X , and a map $\tau : C \rightarrow \mathbb{R}$ that is monotone for $\preceq$ and nonnegative on positive elements, with $\tau(A_X) = 0$ only if A_X equals the zero element.
- (ii') *A link between the two.* If A_X is the zero element, then $\psi_-(x) = 0$ at every visible x .
- (iii) *Domination records of vanishing trace.* Positive elements $B_n \in C$ with $A_X \preceq B_n$ for all n and $\tau(B_n) \rightarrow 0$.
- (iv) *A bridge to the zeros.* For each represented zero ρ of $\text{Sel}_{\zeta, \text{tr}}^!$, a visible object $x_\rho \in X$, and a map $e : R \rightarrow Y$ on the shell's coordinates with $e(r) = 0$ only if $r = 0$, such that

$$\psi_-(x_\rho) = e\left(c(\rho) - \frac{1}{2}\right).$$

Items (i)–(iii) are an instance of the duality-confinement setting in its abstract form: the domination records and the trace squeeze force A_X to vanish, and item (ii') then gives $\psi_- = 0$ at every visible object. None of this involves the zeta function, and neither $A_Z(\zeta)$ nor q appears in the hypothesis; the zeros enter only through item (iv), which asserts that the auxiliary readout at the object attached to ρ encodes the displacement coordinate $c(\rho) - \frac{1}{2}$. In this abstract form the link between A_X and ψ_- is part of the data (item (ii')) rather than a consequence of a construction, and the next proposition shows that the hypothesis is exactly as strong as the critical-line statement for the represented zeros. The operator form, in which A_X is the Gram operator of the readout, is Theorem 7.4. Nothing in the hypothesis requires the auxiliary ledger to be constructed from zeta-function data, and its intended provenance—an instance of the companion paper's structural law obtained from a trace formula—is not established here.

The next proposition measures the strength of the hypothesis.

Proposition 6.2 (Calibration of the recognition source). *For every shell $\text{Sel}_{\zeta, \text{tr}}^!$, $\Gamma_{\text{SDTC-Selberg}}$ holds for $\text{Sel}_{\zeta, \text{tr}}^!$ if and only if $c(\rho) = \frac{1}{2}$ for every represented zero ρ . In particular, $\Gamma_{\text{SDTC-Selberg}}$ holds for the bare shell if and only if the Riemann hypothesis holds.*

Proof. If $\Gamma_{\text{SDTC-Selberg}}$ holds, Theorem 7.2 below gives $c(\rho) = \frac{1}{2}$ for every represented zero. Conversely, suppose $c(\rho) = \frac{1}{2}$ for all represented zeros. Take X a single visible point with J the identity, $Y = \mathbb{R}$, $\psi_- = 0$, $C = \mathbb{R}$ with the usual order and positive elements $[0, \infty)$, $A_X = 0$, $B_n = 0$, τ the identity, x_ρ the point, and $e : R \rightarrow \mathbb{R}$ with $e(0) = 0$ and $e(r) = 1$ for $r \neq 0$ (a classical case distinction on the abstract set R). Items (i)–(iii) hold trivially, $e(r) = 0$ only for $r = 0$, and item (iv) holds because $c(\rho) - \frac{1}{2} = 0$ by the coordinate laws, so $e(c(\rho) - \frac{1}{2}) = 0 = \psi_-(x_\rho)$. On the bare shell, $c(\rho) = \text{Re } \rho$ and the represented zeros are the nontrivial zeros, so the condition is the Riemann hypothesis (Theorem 5.1). $\square$

Thus the recognition source, as a type of data, is exactly as strong as the critical-line statement it is used to prove, and on the bare shell establishing it is no easier than proving the Riemann hypothesis. Its intended content is not its logical strength but its provenance: the conditional theorem becomes a proof only if the data are produced by an independent argument, for instance from a trace formula, in which item (iv) is a consequence rather than an assumption. Theorem 7.4 shows that the same calibration holds for a non-degenerate version built from the actual zeros.

7 The conditional landing chain

This section proves the conditional theorems and the concrete calibration. The reader may keep the bare shell in mind throughout: for it $c(\rho) = \operatorname{Re} \rho$, the represented zeros are all nontrivial zeros, and Theorems 7.2 and 7.3 say the same thing. The duality-confinement theorem is used in the abstract form of the companion paper [29, Remark 7.2]: if a is a positive element of an ordered set, τ is a nonnegative monotone real trace with $\tau(a) = 0$ only if $a = 0$, and $a \preceq b_n$ with $\tau(b_n) \rightarrow 0$, then $a = 0$, because $0 \leq \tau(a) \leq \tau(b_n) \rightarrow 0$.

Theorem 7.1 (Duality-confinement theorem applied to $\operatorname{Sel}_{\zeta, \operatorname{tr}}^!$). *Let $\operatorname{Sel}_{\zeta, \operatorname{tr}}^!$ be a shell for which $\Gamma_{\operatorname{SDTC-Selberg}}$ holds (Hypothesis 6.1). Then the anti-invariant ledger of the represented zeros vanishes: $A_Z(\zeta) = 0$.*

Proof. By items (ii) and (iii) and the abstract squeeze, A_X is the zero element; by item (ii'), $\psi_-(x) = 0$ at every visible object x . For a represented zero ρ , the object x_ρ is visible, so by item (iv) $e(c(\rho) - \frac{1}{2}) = \psi_-(x_\rho) = 0$, and hence $c(\rho) = \frac{1}{2}$. (Separation, item (i), gives the equivalent statement that x_ρ is fixed by J ; it is not needed for the conclusion.) Every summand of $A_Z(\zeta)$ therefore vanishes, and $A_Z(\zeta) = 0$ by Theorem 5.3 applied to the represented zeros. $\square$

The companion theorem enters this proof only through the abstract squeeze; the link from A_X to the readout is item (ii') and the link to the zeros is item (iv), both of which are assumed.

Theorem 7.2 (Conditional landing chain). *Let $\operatorname{Sel}_{\zeta, \operatorname{tr}}^!$ be a shell for which $\Gamma_{\operatorname{SDTC-Selberg}}$ holds. Then every represented zero ρ of $\operatorname{Sel}_{\zeta, \operatorname{tr}}^!$ satisfies $c(\rho) = \frac{1}{2}$. On the bare shell this says $\operatorname{Re} \rho = \frac{1}{2}$ for every nontrivial zero ρ .*

Proof. Theorem 7.1 gives $A_Z(\zeta) = 0$, and the forward direction of Theorem T (Theorem 5.2), applied to the represented zeros, gives the conclusion. $\square$

Theorem 7.3 (Full classical conditional Riemann hypothesis). *Let $\operatorname{Sel}_{\zeta, \operatorname{tr}}^!$ be a shell with coordinate system R , and suppose that $\Gamma_{\operatorname{SDTC-Selberg}}$ holds for $\operatorname{Sel}_{\zeta, \operatorname{tr}}^!$ and that there is a classical-zero identification: a map $t : R \rightarrow \mathbb{R}$ with $t(\frac{1}{2}) = \frac{1}{2}$ such that for every nontrivial zero s of ζ there is a represented zero ρ with $t(c(\rho)) = \operatorname{Re} s$. Then the Riemann hypothesis holds. For the bare shell, t is the identity.*

Proof. By Theorem 7.2, $c(\rho) = \frac{1}{2}$ for every represented zero, so $\operatorname{Re} s = t(\frac{1}{2}) = \frac{1}{2}$ for every nontrivial zero s . The Riemann hypothesis follows, because every zero of ζ other than the trivial zeros lies in the strip $0 < \operatorname{Re} s < 1$ (proof of Theorem 5.1). $\square$

Taken together with Proposition 6.2, these theorems say that the Riemann hypothesis is equivalent to the statement that $\Gamma_{\operatorname{SDTC-Selberg}}$ holds for the bare shell. The recognition source used there is degenerate: a single object carries the readout. The next theorem applies the operator form of the duality-confinement theorem to the actual zeros, with the zeros themselves as objects. Let $(W_n)_{n \geq 1}$ be finite sets of nontrivial zeros, each closed under J_L , such that every nontrivial zero lies in W_n for all large n ; such a sequence exists because the nontrivial zeros are countable and each orbit of J_L has one or two elements (take W_n to be the union of the orbits of the first n zeros in an enumeration). On W_n take the finite weighted ledger of the companion paper with unit weights, response space $\mathbb{R}$, response involution $v \mapsto -v$, involution J_L and readout $\psi_-(\rho) = \operatorname{Re} \rho - \frac{1}{2}$. Its operator is multiplication by the unit-weight window sum $A_{W_n} = \sum_{\rho \in W_n} (\operatorname{Re} \rho - \frac{1}{2})^2$, and its readout is separating: it vanishes exactly at the fixed points of J_L . The response space is one-dimensional, so the operators are scalars and the theorem is the simplest case of the duality-confinement theorem; what it adds to Proposition 6.2 is that the objects are the actual zeros and the bridge is an identity rather than an assumption.

Theorem 7.4 (Concrete window form). *With the windows W_n above, the following are equivalent:*

- (a) *for every n there are real numbers $b_{n,k} \geq A_{W_n}$ with $b_{n,k} \rightarrow 0$ as $k \rightarrow \infty$;*
- (b) *there are real numbers $b_n \geq A_{W_n}$ with $b_n \rightarrow 0$;*
- (c) *the Riemann hypothesis.*

In (a) and (b) the numbers b are the traces of positive operators on $\mathbb{R}$ dominating the window operators, so (a) is the hypothesis of the duality-confinement theorem for each window, and the implication (a) $\Rightarrow$ (c) is proved by that theorem.

Proof. (a) $\Rightarrow$ (c): for each n , $0 \leq A_{W_n} \leq b_{n,k} \rightarrow 0$, so $A_{W_n} = 0$; this is the duality-confinement theorem for this finite weighted ledger [29, Theorem 7.1], and since the readout is separating every zero in W_n is fixed by J_L , that is, lies on the critical line; every nontrivial zero lies in some window, and Theorem 5.1 gives the Riemann hypothesis. (b) $\Rightarrow$ (c): if some nontrivial zero ρ_0 were off the line, it would lie in W_n for all large n , so $b_n \geq A_{W_n} \geq (\operatorname{Re} \rho_0 - \frac{1}{2})^2 > 0$ for all large n , contradicting $b_n \rightarrow 0$. (c) $\Rightarrow$ (a) and (c) $\Rightarrow$ (b): under the Riemann hypothesis every window sum is zero, and the zero numbers dominate it. $\square$

Unwound, condition (a) says that every window sum is zero and condition (b) that the window sums tend to zero; the theorem makes explicit that, for the actual zeros, domination records of vanishing trace are nothing more than such upper bounds, and that they exist exactly when the window sums vanish. A proof of the Riemann hypothesis along this route therefore needs an independent source of such bounds; Section 9 examines some candidate sources.

8 A classification of the hypotheses

A companion paper on audited operational realisability (AOR) [23] provides a vocabulary for classifying the ingredients of an argument: each ingredient receives a *residual type* describing what kind of gap it closes (a missing source, a transport between settings, a presentation choice, a meta-level condition, and so on) and a *status* describing how the gap is closed (by construction, by a supplied bridge, by another paper, or not at all). This section records such a classification for the argument of Section 7. It is bookkeeping: it proves nothing about the zeros, can be skipped by a reader interested only in the mathematics, and does not establish membership of the argument in the stronger, proof-carrying notion of realisability of that paper.

Definition 8.1 (The AOR instance object attached to $\operatorname{Sel}_{\zeta, \operatorname{tr}}^!$). Let $\mathfrak{S}_{\operatorname{RH}}$ be the scope consisting of the shell $\operatorname{Sel}_{\zeta, \operatorname{tr}}^!$, the objects of Section 3, the recognition source $\Gamma_{\operatorname{SDTC-Selberg}}$, the duality-confinement theorem and Theorem T. The AOR instance object $C_{\operatorname{RH}}(\operatorname{Sel}_{\zeta, \operatorname{tr}}^!, \Gamma_{\operatorname{SDTC-Selberg}})$ is the list of the following entries, each with a residual type and a status: the carrier (the zero side of $\operatorname{Sel}_{\zeta, \operatorname{tr}}^!$ with its quotients); the observations ($Z_{\zeta}^{\operatorname{nt}}$, ψ_- , $A_Z(\zeta)$, Vis_L); the routes (J_L , π_L , and the bridge from the auxiliary ledger to the shell); the sources (Audit_L , $\Gamma_{\operatorname{SDTC-Selberg}}$ and its records B_n); the constraints (positivity, order, trace monotonicity, domination, the trace limit); the records 8.2 to 8.8 below; and the non-claims of Section 10.

Each record below is a short list of *atoms*. An atom consists of a *primary* residual type (the kind of gap the entry closes), a list of *forced secondary* types (further kinds of gap that the entry is declared to bring with it, each of which must be recorded in its own right), and a status. The statuses used are: *by construction* (a definition), *zero* (proved, so that no gap remains), *bridged* (supplied as a hypothesis), *asymptotically budgeted* (a limit supplied as a hypothesis), *proved in*

another paper, *outside scope*, and *non-claim*. Theorem 8.9 below is a consistency check on these lists: it says that every kind of gap an entry declares is itself recorded by some entry, and nothing more.

Record 8.2 (Mechanical records). The mechanical record has six atoms, one for each of Definitions 3.1 to 3.4 and 4.1 and Theorem 5.1. The five definition atoms have primary types role (involution), role (readout), presentation (zero ledger), presentation (anti-invariant ledger) and meta (shell), each with status by construction and no forced secondaries; the atom for Theorem T has primary type target, forced secondary transport, and status zero. None uses recognition-source content.

Record 8.3 (Recognition-source record). $\Gamma_{\text{SDTC-Selberg}}$ is recorded by an atom with primary type source, forced secondaries target, role and limit, and status bridged: it is supplied, not derived. Three further atoms record the secondaries: target and role, bridged, and limit, asymptotically budgeted.

Record 8.4 (Bridge-field record). The bridge between the auxiliary ledger and the shell (item (iv) of Hypothesis 6.1 with the readout-coherence fields) is recorded by transport atoms—one with forced secondaries role and target and status zero, two bridged—and bridged atoms for role and target.

Record 8.5 (Admissibility-audit record). The audit marker Audit_L is recorded by one atom of primary type meta and status bridged. Admissibility of an intended trace-formula realization under the framework’s formation rules is not established.

Record 8.6 (Cross-paper record). The use of the duality-confinement theorem of Tsiokos [29] is recorded by an atom of primary type transport with forced secondaries source, role and target and status proved in another paper, and bridged atoms for source, role and target.

Record 8.7 (Real-coordinate record). The abstract real coordinate of a general shell is recorded by two atoms of primary type presentation, with statuses outside scope and non-claim. Its identification with $\text{Re } \rho$ is the separate hypothesis of Theorem 7.3; for the bare shell it holds by construction.

Record 8.8 (Translation record). Theorem T is recorded by an atom of primary type target with forced secondary transport, and an atom of primary type transport, both with status zero.

Theorem 8.9 (Local register completeness). *In the concatenation $C_{\text{RH}}(\text{Sel}_{\zeta, \text{tr}}^1, \Gamma_{\text{SDTC-Selberg}})$ of Records 8.2 to 8.8, the list of non-claims is nonempty, and every forced secondary type of every atom occurs as the primary type of some atom. In the terminology of Tsiokos [23], the register is locally complete on $\mathfrak{S}_{\text{RH}}$.*

Proof. The non-claim list is nonempty by construction. The forced secondary types that occur are transport, target, role, limit and source. Each occurs as a primary type: transport in the bridge-field record, target, role and limit in the recognition-source record, and source in the recognition-source and cross-paper records. $\square$

Remark 8.10 (Partial status of the classification). Theorem 8.9 is a statement about a finite list of labels; it uses no domination record, bridge or trace limit, and it does not imply the critical-line statement. The proof-carrying notion of realisability of Tsiokos [23] requires stored proofs for each obligation and quantitative payment hypotheses, which are not supplied here; membership of the argument in that notion remains open.

9 Finite windows

By Theorem 7.4, a proof along the duality-confinement route needs upper bounds tending to zero for the window sums A_W . This section records some elementary facts about natural candidates for such bounds. The main point is Proposition 9.1(b): the symmetric probe $L_W(x) = \sum x_\rho$ sees nothing of the odd displacement vector, by symmetry alone, so its residual charge is the whole window sum; the equivalences of Proposition 9.3 are then tautologies forced by this choice of probe, and carry no information about ζ . Only the oddness of the displacement under the functional-equation map is used here; no fixed locus is involved.

Setting. Throughout, $\sigma(\rho) = 1 - \rho$ is the functional-equation map on the nontrivial zeros (not to be confused with the reflection J_L in the critical line), a finite set W of nontrivial zeros is *reflected* if $\sigma(W) = W$, and the positive integer weights satisfy $m_{\sigma(\rho)} = m_\rho$. A sequence (W_n) of finite sets is *exhaustive* if every nontrivial zero lies in W_n for all large n ; exhaustive sequences of reflected windows exist because the nontrivial zeros are countable and each orbit of σ has at most two elements. These windows differ from those of Theorem 7.4, which are closed under $J_L(\rho) = 1 - \bar{\rho}$ and use unit weights; windows closed under both reflections also exist, since the two reflections commute and each orbit of the group they generate has at most four elements. On the Euclidean space $\mathbb{R}^W$ put

$$v_W(\rho) = \sqrt{m_\rho} \left(\operatorname{Re} \rho - \frac{1}{2} \right), \quad L_W(x) = \sum_{\rho \in W} x_\rho,$$

the *displacement vector* and the *invariant probe*. For a linear map L from $\mathbb{R}^W$ to a finite-dimensional real inner product space, let L^+ be its Moore–Penrose pseudoinverse and

$$\Xi(L) = I - L^+ L,$$

the orthogonal projection onto $\ker L$; $L^+ L$ is the orthogonal projection onto $(\ker L)^\perp$. For a vector v , the *residual charge* $\langle v, \Xi(L)v \rangle = \|\Xi(L)v\|^2$ measures the part of v that the probe L does not see. We call LL^* the *native currency* of L . (These are the identity-audit cases of the residual, native currency and decoder L^+ of the companion paper [30].) A *payment* for a window sum is an upper bound for it, or for a quadratic form dominating it, that one hopes to derive independently. Operator inequalities $S \preceq T$ are in the sense of quadratic forms.

Proposition 9.1 (Window identities). *Let W be a reflected window with symmetric positive weights.*

- (a) $\|v_W\|^2 = A_W$, and $A_W \leq A_Z(\zeta)$.
- (b) $L_W v_W = 0$. Consequently $\langle v_W, \Xi(L_W)v_W \rangle = A_W$: the invariant probe sees nothing of the displacement, and the residual charge is the whole window sum.
- (c) For every linear map L on $\mathbb{R}^W$, $\langle v_W, (L^+(LL^*)(L^+)^* + \Xi(L))v_W \rangle = A_W$. Changing the probe moves charge between the decoded native currency and the residual, but never changes their total.
- (d) Let P_W be the orthogonal projection onto $\ker L_W$, a probe determined by the window alone (it does not depend on the displacements). Then $\langle v_W, \Xi(P_W)v_W \rangle = 0$ and $\langle v_W, P_W P_W^* v_W \rangle = A_W$. In particular, a window containing a zero off the critical line has zero residual charge for this probe and strictly positive native currency.

Part (d) shows that residual charge depends on the probe; the equivalences of Proposition 9.3 concern the invariant probe L_W .

Proof. (a) is the definition of v_W and of the extended sum. (b): the reflection pairs ρ with $1 - \rho$, which has the opposite displacement and the same weight; the only fixed point $\rho = \frac{1}{2}$ has displacement zero. So the terms of $L_W v_W$ cancel, v_W lies in $\ker L_W$, and $\Xi(L_W)v_W = v_W$. (c): $L^+(LL^*)(L^+)^* = L^+L$ is the orthogonal projection onto $(\ker L)^\perp$, so the operator in (c) is the identity. (d): $\ker P_W = (\ker L_W)^\perp$, which is orthogonal to v_W by (b); and $P_W P_W^* = P_W$ fixes v_W . $\square$

Proposition 9.2 (Two elementary limitations). *Let W be a nonempty window.*

- (a) Operator transport bounds for the identity target cannot vanish. *This part does not use the zeros at all, and the same argument works for any finite-dimensional target of L . Let $L : \mathbb{R}^W \rightarrow \mathbb{R}$ be linear, and suppose $LL^* \preceq \Theta$, $\Xi(L) \preceq \Omega$ and $L^+\Theta(L^+)^* + \Omega \preceq H \cdot I$ for operators Θ, Ω and a real number H . Then $H \geq 1$.*
- (b) The probe cannot be coercive on its own residual. *If $c\|x\| \leq |L_W x|$ for some $c > 0$ and all x in the range of $\Xi(L_W)$, then L_W is injective. Hence no such c exists when W has at least two elements.*

Proof. (a) Conjugating $LL^* \preceq \Theta$ by L^+ and adding $\Xi(L) \preceq \Omega$ gives $I = L^+L + \Xi(L) \preceq L^+\Theta(L^+)^* + \Omega \preceq H \cdot I$, and $\mathbb{R}^W \neq 0$. (b) The range of $\Xi(L_W)$ is $\ker L_W$, on which L_W vanishes; the inequality forces $\ker L_W = 0$. A nonzero linear functional on $\mathbb{R}^W$ is injective only if $\dim \mathbb{R}^W = 1$. $\square$

Here the *identity target* is the map I on $\mathbb{R}^W$ whose values the payment is meant to control, and a *transport bound* is the scalar H in an inequality of the form in (a), which dominates the decoded native budget plus the residual bound by $H \cdot I$. Part (b) says that coercivity on the residual forces L_W to be injective, which fails as soon as W has at least two elements, because then $\ker L_W \neq 0$. Part (a) says only that the identity operator is not small: a transport bound of the form in (a), which dominates the identity target as an operator rather than on the actual displacement vector v_W , cannot tend to zero.

Proposition 9.3 (Payments that restate the Riemann hypothesis). *Let (W_n) be an exhaustive sequence of reflected windows with symmetric positive weights, and write $v_n = v_{W_n}$, $L_n = L_{W_n}$, $A_n = A_{W_n}$. Each of the following is equivalent to the Riemann hypothesis:*

- (a) $\langle v_n, \Xi(L_n)v_n \rangle = 0$ for every n ;
- (b) there are $b_n \rightarrow 0$ with $\langle v_n, \Xi(L_n)v_n \rangle \leq b_n$;
- (c) there are operators $\Omega_n \succeq \Xi(L_n)$ and $b_n \rightarrow 0$ with $\langle v_n, \Omega_n v_n \rangle \leq b_n$;
- (d) for a fixed real number $q < 1$, $\langle v_n, \Xi(L_n)v_n \rangle \leq q A_n$ for every n ;
- (e) $\sum_n A_n < \infty$.

Proof. By Proposition 9.1(b), the residual charge in (a), (b) and (d) equals A_n , and in (c) it is at most $\langle v_n, \Omega_n v_n \rangle$. If some zero ρ_0 is off the critical line, it lies in W_n for all large n , and then $A_n \geq (\operatorname{Re} \rho_0 - \frac{1}{2})^2 > 0$ for all large n ; this contradicts each of (a)–(e) (for (d), $A_n \leq q A_n$ forces $A_n = 0$; for (e), the terms of a convergent series tend to zero). Conversely, under the Riemann hypothesis every A_n is zero, and (a)–(e) hold with $\Omega_n = \Xi(L_n)$ and $b_n = 0$. $\square$

The proposition does not say that the Riemann hypothesis is hard; it says that, on actual zero windows, bounding the residual charge, bounding any operator above it, asking for a uniform residual gap, or asking for summable window sums is each the Riemann hypothesis in another form. An argument along these lines becomes a proof only if the bound is derived from zeta-function data by a route that does not already presuppose it.

10 Scope and non-claims

No unconditional claim. No statement about the location of the zeros of ζ is proved. The conditional theorems assume $\Gamma_{\text{SDTC-Selberg}}$, and Theorem 7.3 also assumes a classical-zero identification. By Proposition 6.2 and Theorem 7.4, on the bare shell of all nontrivial zeros with their real parts, and for windows that eventually contain every zero, these hypotheses, as types of data, are equivalent to the Riemann hypothesis; the conditional theorems are reformulations and provide no evidence for it.

The recognition source is not constructed. The intended provenance of $\Gamma_{\text{SDTC-Selberg}}$ —a trace formula of Selberg type for the zeros of ζ , realizing the structural law of the companion paper—is not established, and no trace identity between spectral and arithmetic data is used. The shell’s name records that intention only (Section 4).

Admissibility is not established. The framework’s formation rules [25] would require an intended trace-formula realization of the shell to be admissible; this is not verified, and the shell’s audit marker is a label, not a proof.

The classification is bookkeeping. The register of Section 8 is a finite list of labels. Its completeness (Theorem 8.9) does not imply the critical-line statement, and the stronger, proof-carrying notion of realisability of Tsiokos [23] is not established for this argument.

Other L -functions and finer questions. Only the Riemann zeta function is treated. Nothing is claimed about the generalized Riemann hypothesis for Dirichlet, modular or other L -functions of the Selberg class [21, 13], about simplicity of the zeros [17, 6], or about zero density [16, 11].

11 Discussion

What the route requires. The duality-confinement route reduces the Riemann hypothesis to upper bounds tending to zero for the window sums A_W , or for operators dominating the associated ledgers. For windows closed under the reflection $s \mapsto 1 - \bar{s}$ (Theorem 7.4), or under the map $s \mapsto 1 - s$ with invariant weights (Proposition 9.3), and along windows that eventually contain every zero, the existence of such bounds is the Riemann hypothesis itself, whatever form the bounds take among those considered. What would be new is a source of bounds: an argument, for instance from an explicit formula or a trace formula, that produces them from zeta-function data without presupposing the conclusion. Proposition 9.2 records the elementary fact that an operator-level transport bound for the identity target, of the form stated there, cannot tend to zero on a nonempty window, so a payment of that kind must be evaluated on the actual displacement vector.

Relation to classical programs. Several classical approaches seek a positivity statement equivalent to the Riemann hypothesis: Weil’s criterion, which asks for positivity of a quadratic functional built from the explicit formula [31, 4], and Li’s criterion [14]; the Nyman–Beurling criterion [18, 3, 1]; de Branges’ spaces of entire functions [7, 8]; and spectral interpretations of the zeros [2, 5]. The window sums of this paper are also positivity data, but of a much cruder kind: they are the squared displacements themselves, and, unlike Weil’s functional, which is built from the explicit formula relating zeros to primes, they involve no primes. A bound for them that did

not presuppose the Riemann hypothesis would presumably have to come from a deeper source of this kind; nothing in the present paper supplies one.

Relation to the companion papers. The duality-confinement theorem is proved in Tsiokos [29]; this paper is its application to the zeros of ζ , and Theorem 7.4 is the zeta-specific case of the converse proved there. The residual operator of Section 9 is the identity-audit case of the adequacy residual of Tsiokos [30], which in the companion Navier–Stokes paper [28] is applied to a physical target; the metatheory paper [27] relates the two applications through a common Gaussian observation. None of these connections supplies a payment for either problem.

Further directions. Three questions remain open: to construct a recognition source from a trace formula, which would turn Theorem 7.3 into a proof; to verify the framework’s admissibility conditions for such a construction; and to treat other L -functions, each of which needs its own ledger and source.

12 Conclusion

The Riemann hypothesis is the vanishing of the positive ledger $A_Z(\zeta) = \sum_{\rho} m_{\rho}(\operatorname{Re} \rho - \frac{1}{2})^2$ of squared displacements of the nontrivial zeros from the fixed line of the involution $s \mapsto 1 - \bar{s}$. If a recognition source supplies domination records of vanishing trace together with a bridge encoding these displacements in an auxiliary readout, the duality-confinement theorem places every represented zero on the critical line of its coordinates, and, with a map from coordinates to the reals that sends $\frac{1}{2}$ to $\frac{1}{2}$ and carries the represented zeros onto the real parts of all nontrivial zeros, the Riemann hypothesis follows. On any shell this hypothesis is equivalent to the critical-line statement for the represented zeros, hence on the bare shell to the Riemann hypothesis, and so is its concrete form on finite windows closed under the reflection $s \mapsto 1 - \bar{s}$, with unit weights, that eventually contain every zero. For windows closed under $s \mapsto 1 - s$, with invariant weights, an operator-level transport bound for the full identity target cannot tend to zero on a nonempty window, and several natural forms of a payment for the window sums along windows that eventually contain every zero are again equivalent to the Riemann hypothesis. These facts are elementary; they use analytic properties of ζ such as the functional equation and the zero-free half-plane $\operatorname{Re} s \geq 1$, but no quantitative or prime-side input. The paper thereby describes what a proof along this route would need, shows that the obvious candidates are no easier than the Riemann hypothesis, and does not claim to supply one. The mathematical statements are formalized in Lean 4 with Mathlib (Appendix A).

A Formal verification

The theorems, propositions and definitions of this paper are formally verified in the Lean 4 proof assistant (version 4.28.0) with the Mathlib library (revision 8f9d9cff6bd728b17a24e163c9402775d9e6a365). The Riemann hypothesis in the conclusions is Mathlib’s predicate `RiemannHypothesis`: every zero of `riemannZeta` that is not a trivial zero and not equal to 1 has real part $\frac{1}{2}$. This appendix describes how the objects of the paper are represented and lists the Lean theorem behind each numbered result.

Representation

The actual nontrivial zeros are the complex numbers s with `completedRiemannZeta s = 0` and $0 < \operatorname{Re} s < 1$; the anti-invariant ledger is a sum in $[0, \infty]$ over them with arbitrary positive natural weights. A shell is a Lean structure whose zero ledger uses an abstract real coordinate; its trace, quotient, visibility and audit fields are arbitrary types without laws, as stated in Section 4. The bare shell instantiates the coordinate as $\mathbb{R}$ and the zero ledger as the actual nontrivial zeros with unit weights. The recognition source $\Gamma_{\text{SDTC-Selberg}}$ is a Lean structure whose fields are the data of Hypothesis 6.1; it is not an axiom, and every theorem that uses it takes an element of it (data witnessing the hypothesis) as an explicit argument. Its auxiliary ledger is expressed in the abstract, library-free interface of the first formulation of the companion paper, which also carries some unused structure (weights, a response involution, a quantitative separation modulus). The formal and the written versions of the hypothesis are equivalent: each holds for a shell exactly when every represented zero has coordinate $\frac{1}{2}$, by Proposition 6.2 and its formal counterpart. The concrete window form (Theorem 7.4) uses instead the Mathlib operator formulation of the companion paper’s Version 3, with positive operators on $\mathbb{R}$ and the linear-algebra trace.

What is not formally verified

The classical facts cited in the text about the functional equation, the location of zeros of ζ outside the critical strip and the countability of the zeros are taken from, or derived from, Mathlib’s formal development of the zeta function wherever the proofs use them. Not verified are: Remark 8.10, which is a scope statement about the stronger notion of realisability; the framework material of Section 2; and the discussion of provenance and admissibility. The identification, stated in Section 9, of the companion paper’s residual and decoder at identity audit with $\Xi(L)$ and L^+ is formally verified (see Proposition 9.1 in the table).

Correspondence

Table 1. Numbered results and the Lean declarations that prove or realize them; names are relative to the namespace `SixBirdsDualityConfinement.RH`.

Result	Lean declarations
Definitions 3.1 and 3.2	<code>ActualInvolution.J</code> , <code>ActualInvolution.J_involutive</code> , <code>ActualInvolution.J_fixed_iff</code> , <code>ActualInvolution.J_zero</code> ; shell form <code>Involution.feInvolution</code> , <code>Involution.psiMinusRh</code>
Definitions 3.3 and 3.4	<code>ClassicalZeroLedger.NontrivialZero</code> , <code>ClassicalZeroLedger.antiInvariantSum</code> ; shell form <code>ZeroLedger.NontrivialZeroLedger</code> , <code>AntiInvariantZeroLedger.AntiInvariantZeroLedger</code>
Definition 4.1	<code>SatSelShell.SatSelShell</code> ; bare shell <code>ClassicalBareShell.bareShell</code>
Theorems 5.1 to 5.3	<code>ClassicalZeroLedger.antiInvariantSum_eq_zero_iff</code> , <code>ClassicalZeroLedger.antiInvariantSum_eq_zero_iff_criticalStripRH</code> , <code>FullClassicalRHBridge.criticalStripRH_iff_riemannHypothesis</code> ; shell form <code>TranslationT.translationT</code> , <code>TranslationT.translationTForward</code> , <code>TranslationT.translationTReverse</code>
Hypothesis 6.1	<code>RHConditional.GammaSdtcSelberg</code> (a structure; its elements are the data (i)–(iv))
Proposition 6.2	<code>BareSourceConverse.gamma_nonempty_iff_shell_criticalLine</code> , <code>BareSourceConverse.gamma_nonempty_iff_criticalStripRH</code> , <code>BareSourceConverse.gammaOfRH</code>
Theorem 7.1	<code>DCMasterApplied.dcMasterAppliedRealTrace</code>

Result	Lean declarations (continued)
Theorem 7.2	RHConditional.rhConditional, RHConditional.rhConditionalClassical
Theorem 7.3	FullClassicalRHBridge.rhConditionalFullClassical, RHConditional.ClassicalZeroIdentification
Theorem 7.4	ConcreteZeroWindowDC.windowRecords_family_iff_riemannHypothesis, ConcreteZeroWindowDC.movingWindowRecords_iff_riemannHypothesis; ConcreteZeroWindowDC.windowPsi_separates, ConcreteZeroWindowDC.weightedLedger_eq_energy, ConcreteZeroWindowDC.windowRecords_iff_fixed
Definition 8.1 and Records 8.2 to 8.8	AORInstance.assembledCarrier, AORInstance.aorMechanicalRecords, AORInstance.aorRecognitionDischarge, AORInstance.aorGammaBridgeDischarge, AORInstance.aorAuditLDischarge, AORInstance.aorDCMasterImportDischarge, AORInstance.aorRealCoordinateDischarge, AORInstance.aorTranslationInterfaceDischarge (literal lists)
Theorem 8.9	AORInstance.aorInstance
Proposition 9.1	FiniteZeroWindows.displacementVector_norm_sq, FiniteZeroWindows.finite_window_le_global_classical_ledger, FiniteZeroWindows.invariantProbe_displacement_zero, FiniteZeroWindows.fixedTarget_window_quadratic_eq_windowEnergy, FiniteZeroWindows.window_complete_currency_eq_windowEnergy, FiniteZeroWindows.blindSubspaceNative_actual_residual_zero, FiniteZeroWindows.blindSubspaceNative_actual_currency_eq_windowEnergy, FiniteZeroWindows.blindSubspaceNative_offLine_control; residual as projection and decoder as pseudoinverse: SixBirdsNeedles.XiCore.auditResidual_id_eq_hilbertKernelXi, SixBirdsNeedles.XiCore.adjoint_pinv_gram_eq_pinv (from the companion Needles library), ConcreteWindowXI.identity_auditDecoder_comp
Proposition 9.2	FiniteZeroWindows.window_fullReadout_transport_floor, FiniteZeroWindows.native_coercive_on_own_residual_forces_injective, ConcreteWindowXI.invariantProbe_not_injective, ConcreteWindowXI.invariantProbe_no_residual_coercivity
Proposition 9.3	FiniteZeroWindows.exhaustive_fixedTarget_xi_zero_iff_criticalStripRH, FiniteZeroWindows.exists_vanishing_fixedTarget_xi_budget_iff_criticalStripRH, FiniteZeroWindows.exists_operatorResidualBudget_iff_criticalStripRH, FiniteZeroWindows.subunit_fixedTarget_xi_gap_iff_criticalStripRH, FiniteZeroWindows.summable_windowEnergy_iff_criticalStripRH; with FullClassicalRHBridge.criticalStripRH_iff_riemannHypothesis

In Section 9 the formal statements use the reflection $\rho \mapsto 1 - \rho$ and positive weights symmetric under it; those for Theorem 7.4 use the involution J_L and unit weights. Statements whose Lean conclusion is the critical-strip form of the Riemann hypothesis are combined with `FullClassicalRHBridge.criticalStripRH_iff_riemannHypothesis`.

Trust base and availability

Every declaration listed depends only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`; no `sorry` and no additional axioms occur. The development is available at <https://github.com/ioannist/six-birds-duality-confinement>; it also contains further finite-window and Gaussian-observation results not used in this paper.

B Version history

Every numbered result of Version 1 keeps its number and place in later versions, and its name except where noted below.

Version 3 (October 1, 2026)

- The paper was rewritten for readers outside the framework; it now says plainly that the conditional theorem is a reformulation of the Riemann hypothesis, not evidence for it. Discussion of the formal development was moved to Appendix A; the numbered lists of non-claims and “no-smuggling gates” and the reference to a “V-Differential” classification, which is not defined in the cited literature, were removed; the account of a three-route derivation attempt for the recognition source was removed.
- Definition 3.3 now allows arbitrary positive weights, with multiplicities as the natural choice; Version 2 required multiplicities, which the formal development does not use. Theorem 5.1 is stated for the actual nontrivial zeros and includes the equivalence with the Riemann hypothesis; the shell version is described in the text.
- Hypothesis 6.1 is now stated in mathematical terms and displayed as a hypothesis (it was labelled “Obligation”). The new Proposition 6.2 states that for every shell it is equivalent to the critical-line statement for the represented zeros, hence on the bare shell to the Riemann hypothesis; Version 2 stated the bare-shell case in prose. Definition 4.1 now states the coordinate system and zero ledger of a shell explicitly.
- The new Theorem 7.4 applies the duality-confinement theorem, in the concrete operator form of the companion paper’s Version 3, to finite reflection-closed windows of actual zeros that eventually contain every zero, and proves that the resulting domination records exist if and only if the Riemann hypothesis holds.
- The AOR classification (Section 8) was condensed. Items 8.2–8.8, which record classification labels and were displayed as theorems, are now displayed as “Records” and describe the recorded lists exactly; Theorem 8.9 now includes the non-emptiness of the non-claim list. The section was renamed “A classification of the hypotheses”.
- The finite-window material of the Version 2 discussion was reorganized into the new Section 9, with the new Propositions 9.1 to 9.3 and complete proofs.
- Names shortened: Definition 4.1 (formerly “Saturated completed Selberg trace shell interface”), Theorem 7.1 (formerly “Duality-confinement master theorem, applied to” the shell), Theorem 7.3 (formerly “Full classical conditional RH”), Records 8.2–8.8 (formerly “Mechanical AOR records”, “Recognition-source discharge”, “Bridge-field discharge”, “Admissibility-audit status record”, “Cross-paper master-theorem discharge”, “Typed real-coordinate discharge”, “Translation-interface discharge”), Theorem 8.9 (formerly “RH local-register theorem”), and Remark 8.10 (formerly “Partial status of the AOR-instance wrapper”).
- Cross-references now name the kind of result they point to; bibliography entries were corrected.

Version 2 (September 28, 2026)

Version 1 described the argument as a refinement-stable audited-operational-realizability instance and identified the shell’s real-part coordinate with the classical real part by construction. Version 2 replaced the first claim by the local-register statement, which is what was proved; made the identification an explicit hypothesis, constructed it for the bare shell, and proved the passage from the critical strip to the full Riemann hypothesis; recorded that on the bare shell the recognition source exists exactly under the Riemann hypothesis; added the finite-window analysis and the comparison with the companion Navier–Stokes and metatheory papers; and moved the formal development from a library-free encoding to Mathlib, correcting the description of what it proves.

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